# HealthConnect Rwanda

Business Insights: patients, clinical activity, billing, operations and patient experience

Data Analytics track | Week 6

CONTEXT AND OBJECTIVES

# Why this analysis matters

- HealthConnect Rwanda needs one view of patients, clinical activity, billing and patient experience.
- The objective: turn 27 source tables into a dashboard and a set of recommendations management can act on.
- Each finding states its basis, so the figures can be checked against the data.

**6**

dashboard pages, each checked against reference figures

**40**

business questions answered in SQL

**27**

source tables loaded and checked

THE DATA

# What the data contains

**20,000**
patients

**60,000**
diagnoses and bills

**5,000**
appointments with clinical detail

**21,064**
patient feedback responses

- Covers customers, appointments, diagnoses, billing, payments, insurance claims, staff, facilities, equipment and feedback.
- Relationships were checked: each billing row links to an appointment, and each payment and claim links to a bill.
- A duplicate facility file was excluded before loading.

DATA LIMITATIONS

# Four limits shape every finding

## Appointment detail

Only 5,000 of 60,000 appointments have facility and department details. Location findings cover that subset.

## Billing has no dates

The monthly trend uses the 5,000 matched bills, totalling 771 million of 9.31 billion billed.

## Missing records

No laboratory, prescription, admission or referral data. Emergency visits are a proxy for referrals.

## Facility names

Facilities are identified by ID only. The master file contains department data.

APPROACH

# How the analysis was built

### 1. Load

PostgreSQL database, 27 tables, checked for row counts and keys

### 2. Query

40 SQL questions, tested and checked against reference values

### 3. Model

Power BI star-style model with 17 measures on the core tables

### 4. Report

Six dashboard pages, each verified, plus a written insights report

PATIENTS

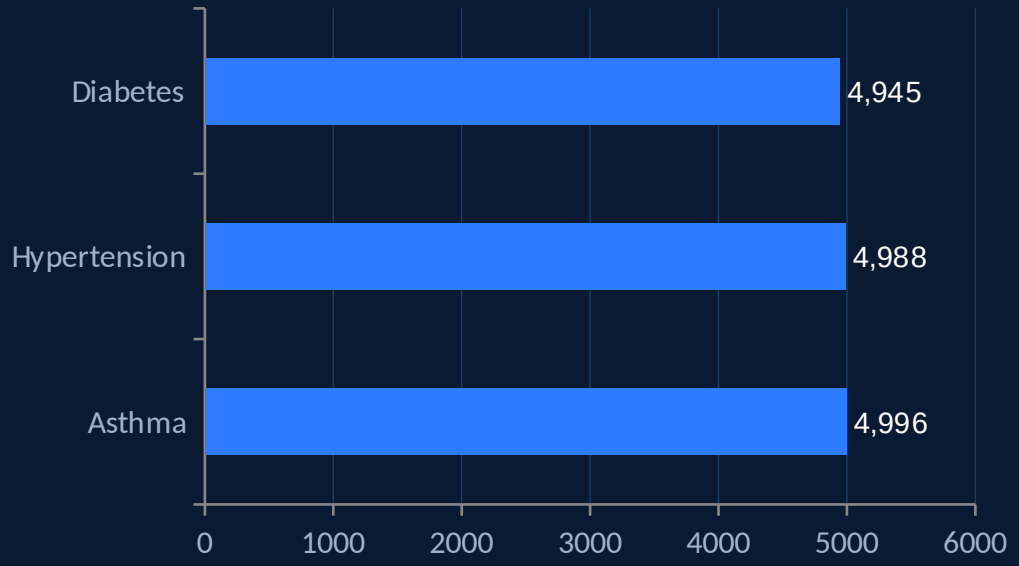
# Patients are split evenly across gender and provinces

**20,000**
registered patients

**45**
average age

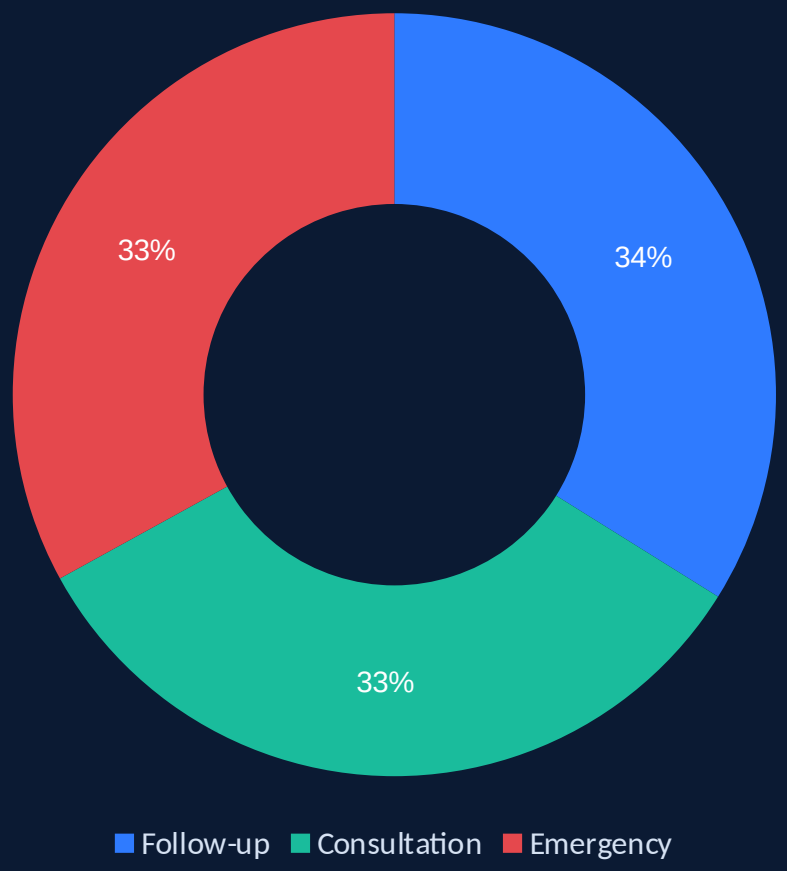
**14,929**
patients with a chronic condition

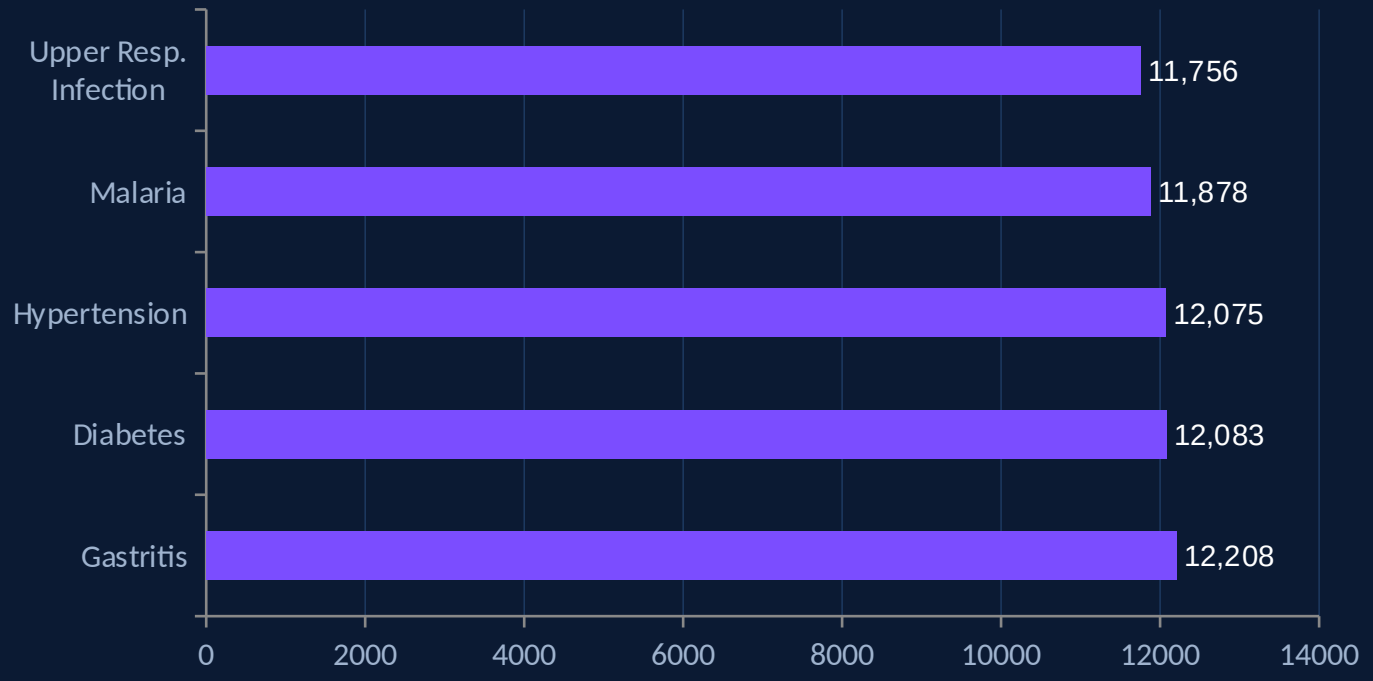
- Gender: 10,090 male and 9,910 female.
- Provinces: between 3,923 and 4,091 patients each, with Western the largest.
- Chronic conditions: asthma 4,996, hypertension 4,988, diabetes 4,945.



CLINICAL ACTIVITY

# Visits and diagnoses are spread evenly

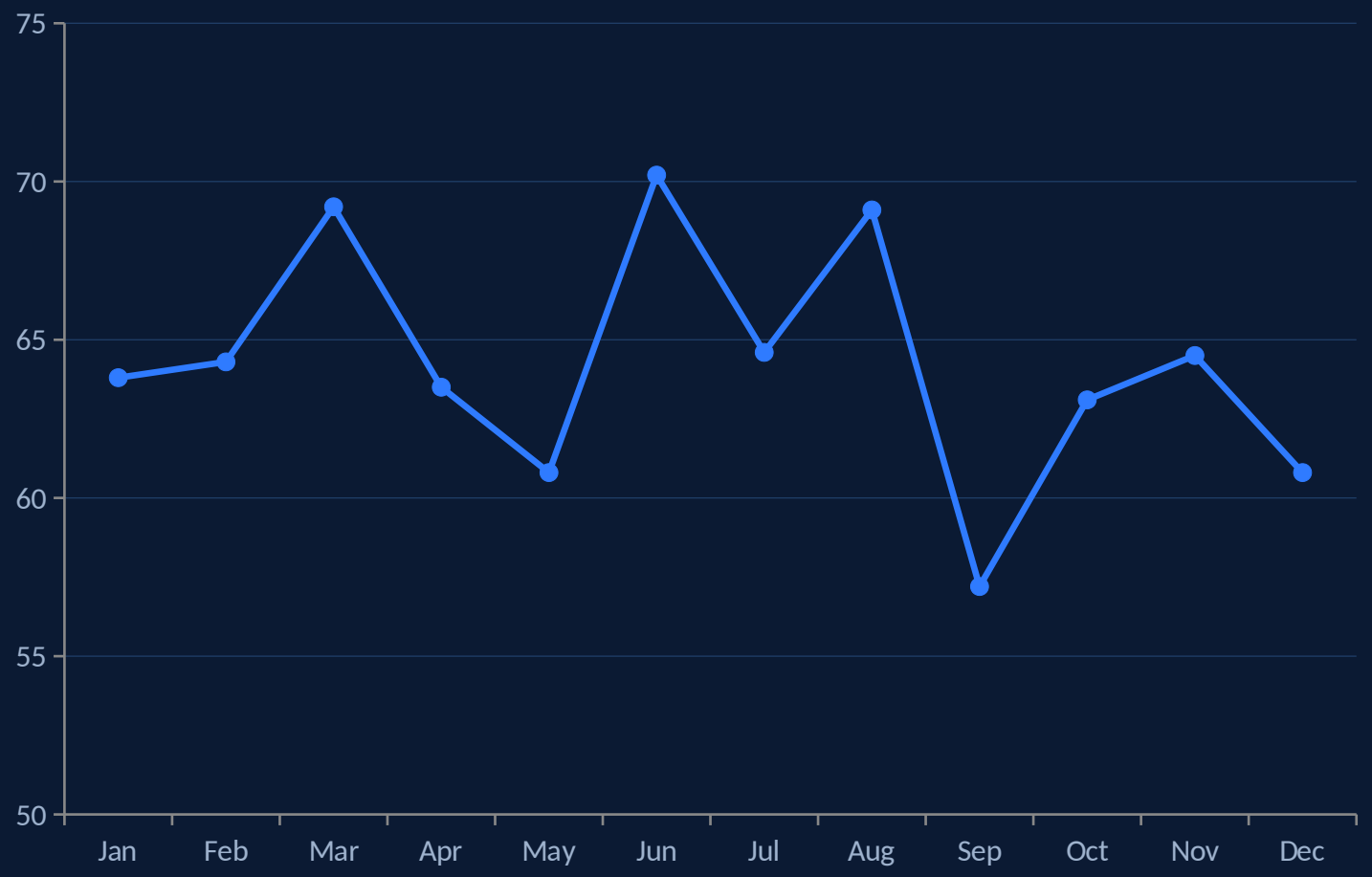



Severity is also balanced: 20,190 mild, 19,827 moderate and 19,983 severe. Emergency visits (1,650) are a proxy for referrals, which are not recorded.

FINANCIAL PERFORMANCE

# Billed revenue holds a steady monthly pattern




**9.31bn**

total billed on all 60,000 bills

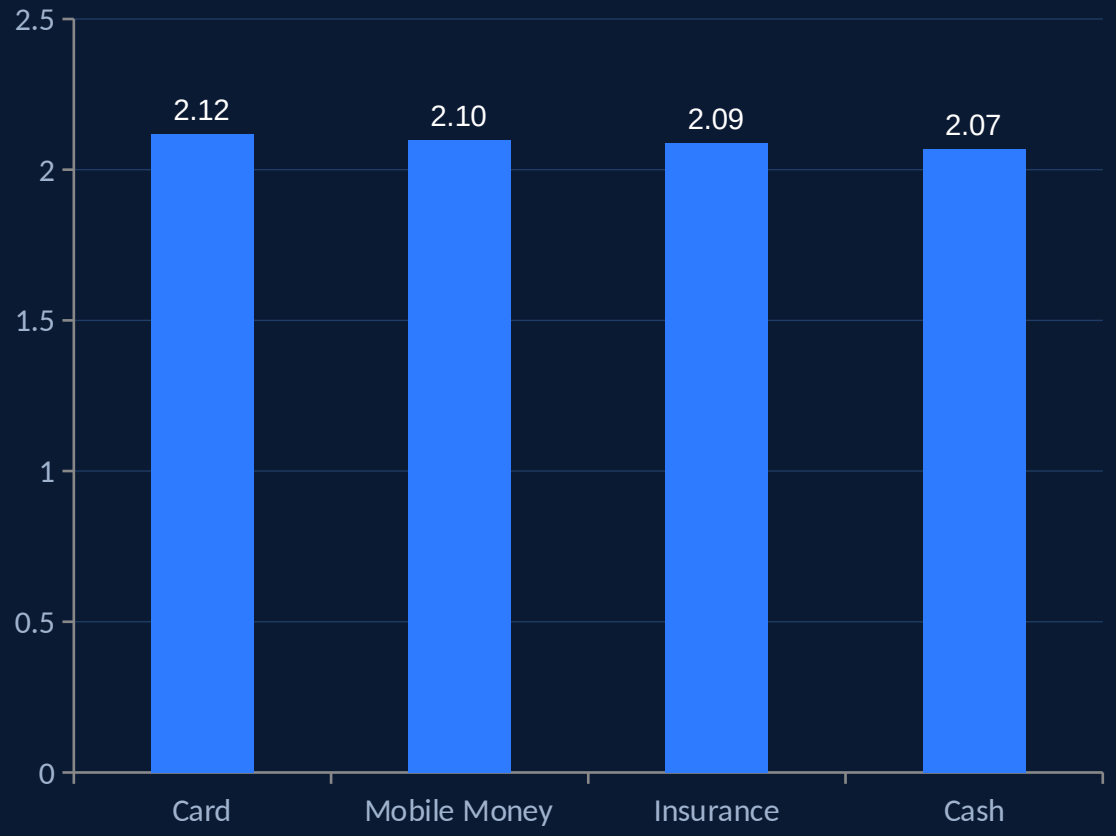
**771M**

billed on the 5,000 matched bills (basis of the trend)

Monthly range: 57.2M (September) to 70.2M (June). No sustained growth or decline.

PAYMENTS AND CLAIMS

# Payments and claims are spread across methods and insurers



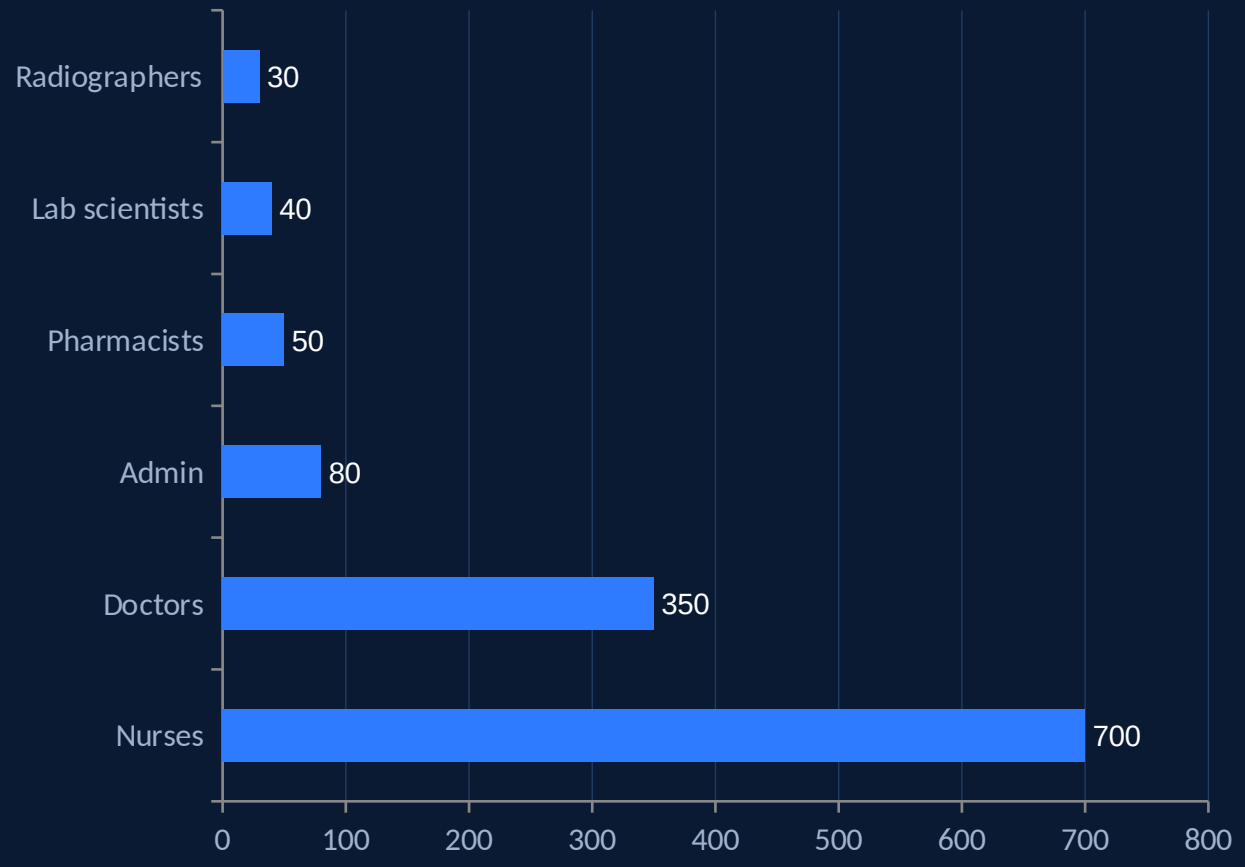




Payments total 8.39bn, and card is the largest method by a small margin. Claims total 4.88bn. Outstanding balance is 919.95M.

OPERATIONS

# Staff and beds are counted, but use is not recorded




**1,250**

staff in total

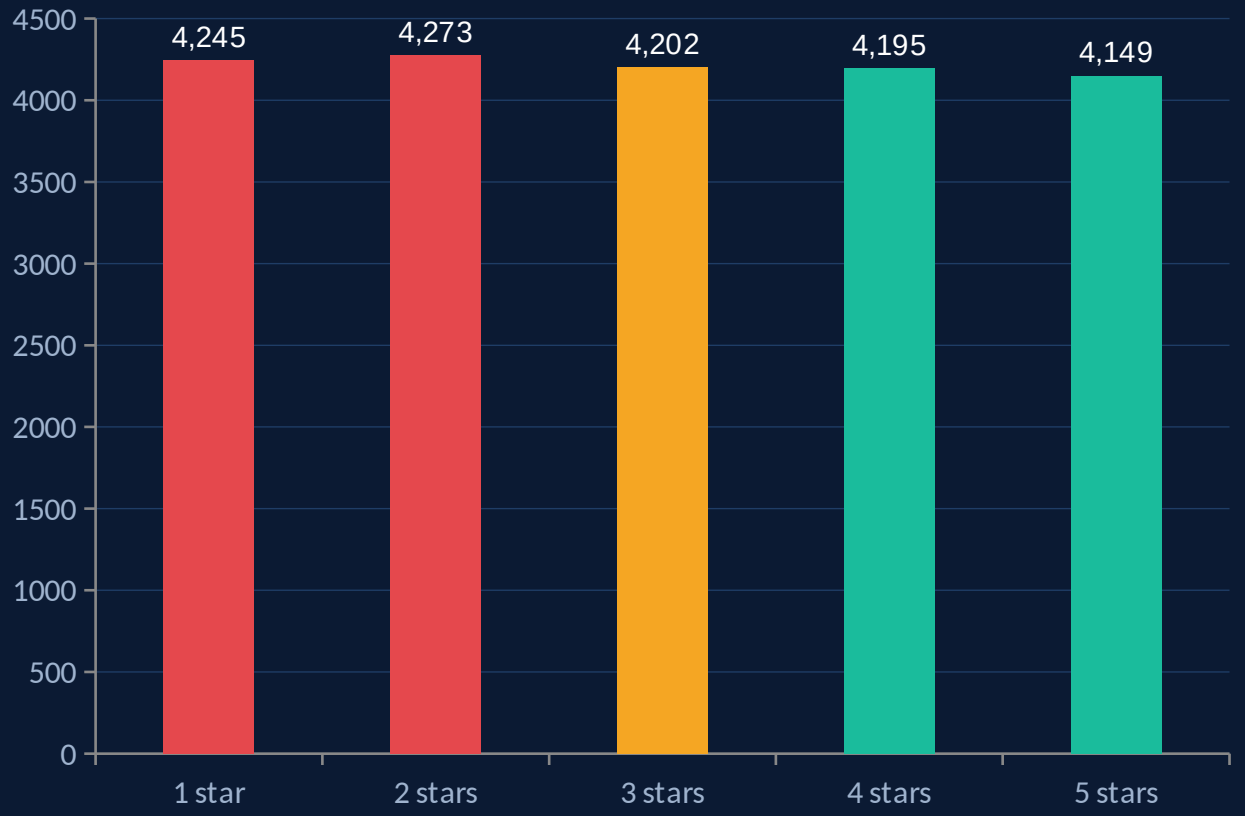
**2,000**

beds across 400 rooms

Every bed is recorded as available, so occupancy cannot be measured. The 200 equipment items across 10 types have no usage data, so utilisation is also unknown.

PATIENT EXPERIENCE

# Four in ten patient responses are complaints




**2.99**

average rating out of 5

**40.4%**

rated 1 or 2 (8,518 responses)

Only 1,756 responses link to a facility, so facility rankings use that subset. HF011 has the highest average at 3.19.

RECOMMENDATIONS

# Five actions to close the gaps

1. **Link every appointment** to its facility and department, so location and revenue findings cover all 60,000 bills.
2. **Date every bill** so revenue can be reported monthly for the full billing base.
3. **Record occupancy and usage** for beds and equipment, so utilisation can be measured.
4. **Review complaints** behind the 40.4% of responses rated 1 or 2, starting with the lowest-rated facilities.
5. **Record referrals, tests, prescriptions and admissions** so the clinical picture is complete.

# Conclusion

- The core operations and patient records are complete and checked.
- Revenue and location findings are strong on the matched subset, and their limit is clearly stated.
- Complaints and missing data are the priorities: linking appointments and recording complaints would give the biggest gain.

*Questions and discussion*